

Building a Circular Economy: How Bagaria Brothers are Powering India's Battery Recycling Revolution

India's lithium-ion battery (LIB) market is witnessing unprecedented expansion, with yearly demand projected to leap from 10.8 GWh in 2022 to an impressive 160.3 GWh by 2030. At present, the country relies heavily on imports, which surged from \$384.6 million in 2019 to \$2.8 billion by 2023. In response, India is aggressively boosting its domestic LIB manufacturing capabilities, aiming to achieve an annual capacity of 150 GWh by 2030, thanks to robust government incentives and increased private sector participation.

To further reduce import dependence and build a resilient circular economy, NavPrakriti Green Energies Pvt Ltd. is pioneering advanced battery recycling and refurbishment solutions. By recovering valuable materials from end-of-life batteries and integrating them back into the domestic supply chain, NavPrakriti is helping to strengthen India's self-sufficiency, support local manufacturing, and ensure sustainable growth for the nation's battery ecosystem.

Founded by brothers Akhilesh and Avnish Bagaria, NavPrakriti's Kolkata-headquartered team inaugurated Eastern India's first large-scale lithium-ion battery recycling and refurbishment facility this year. This milestone is not just a step forward for India's circular economy but also marks a decisive shift for East India, a region long underserved by advanced recycling infrastructure. By establishing a state-of-the-art facility in Kolkata, NavPrakriti is working to ensure that the benefits of a circular economy reach new geographies and communities.

Akhilesh Bagaria, co-founder, brings financial discipline and a passion for sustainability. An alumnus of St. Xavier's College and a Chartered Accountant, Akhilesh further honed his entrepre-

neurial instincts at the University of Warwick. His focus on "urban mining" is central to NavPrakriti's mission: resources, once used, must be reused to preserve them for future generations. The company's rigorous compliance with the National Critical Mineral Mission and Battery Waste Management Rules 2022 ensures operational transparency and best-in-class processes.

Avnish Bagaria, co-founder, adds strategic clarity and a hands-on management style. A Warwick Business School graduate and experienced entrepreneur, Avnish has driven the scaling of operations, development of industry partnerships, and execution of ambitious growth plans. His background in automotive retail and angel investing has helped NavPrakriti attract talent and foster a results-driven culture.

Since its founding in 2024, NavPrakriti has built a robust supply chain, sourcing end-of-life batteries from consumer products, telecom infrastructure, and industrial applications. What sets the company apart is its commitment to closing the loop in the circular economy, especially in East India. By working closely with leading battery manufacturers and OEMs, NavPrakriti is building an integrated ecosystem for collection, recycling, and refurbishment under the EPR (Extended Producer Responsibility) framework. Strategic partnerships with research bodies like CMET Hyderabad have positioned NavPrakriti at the cutting edge of recycling technology.

The facility processes 12,000 tonnes of used batteries annually, with plans to scale up to 24,000 tonnes. Designed to recover critical minerals such as aluminium, copper, nickel, cobalt, manganese, and

lithium, the operation seeks to localize and maximize resource recovery. For the Bagaria brothers, this is more than a business opportunity, it is about building the backbone of India's circular economy and advancing the clean energy transition, with a special focus on East India.

NavPrakriti's roadmap is focused and ambitious. By 2030, the company aims to recover and refine over 95% of battery metals domestically, create India's largest integrated recycling and manufacturing hub, and enable second-life battery applications for energy storage. The company is also strengthening its presence in the untapped East Indian market by empowering local industries and building strategic alliances with manufacturers to truly close the loop of the circular economy.

Looking ahead, NavPrakriti is exploring international partnerships for technology development and knowledge exchange to further enhance its capabilities and bring global best practices to India. However, its primary mission remains clear: to lead the circular economy revolution in East India, set new benchmarks for sustainable industry, and power India's clean energy future.

